

PHYSICS I
(PNAP0123)
SEMESTER I
SESSION 2026/2027

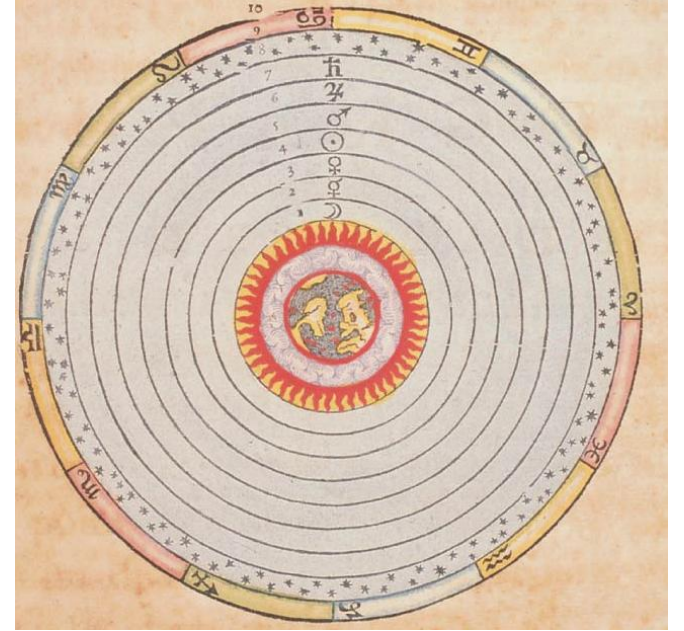
Chapter 1: Units, Measurement, Math Review

Lectured by: Dr. Nurul Izzah Mukri

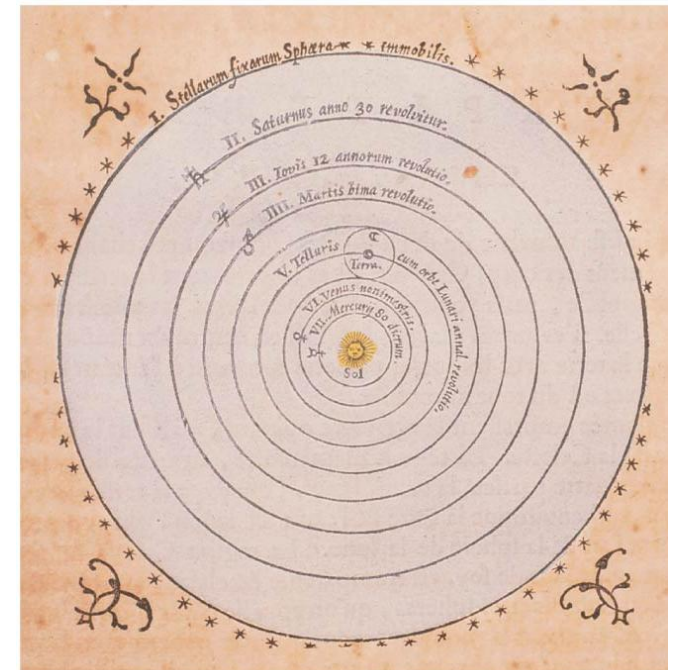
Why Physics is important?

1.1 Introduction

- Observation: important first step toward scientific theory; requires imagination to tell what is important.
- Prediction is created.
- Observations(experimentation and measurements) will tell if the prediction is accurate.
- Theories: created to explain observations. And the cycle goes on



Ptolemy's geocentric view of the universe.



Copernicus's heliocentric view of the universe

Content of Chapter 1:

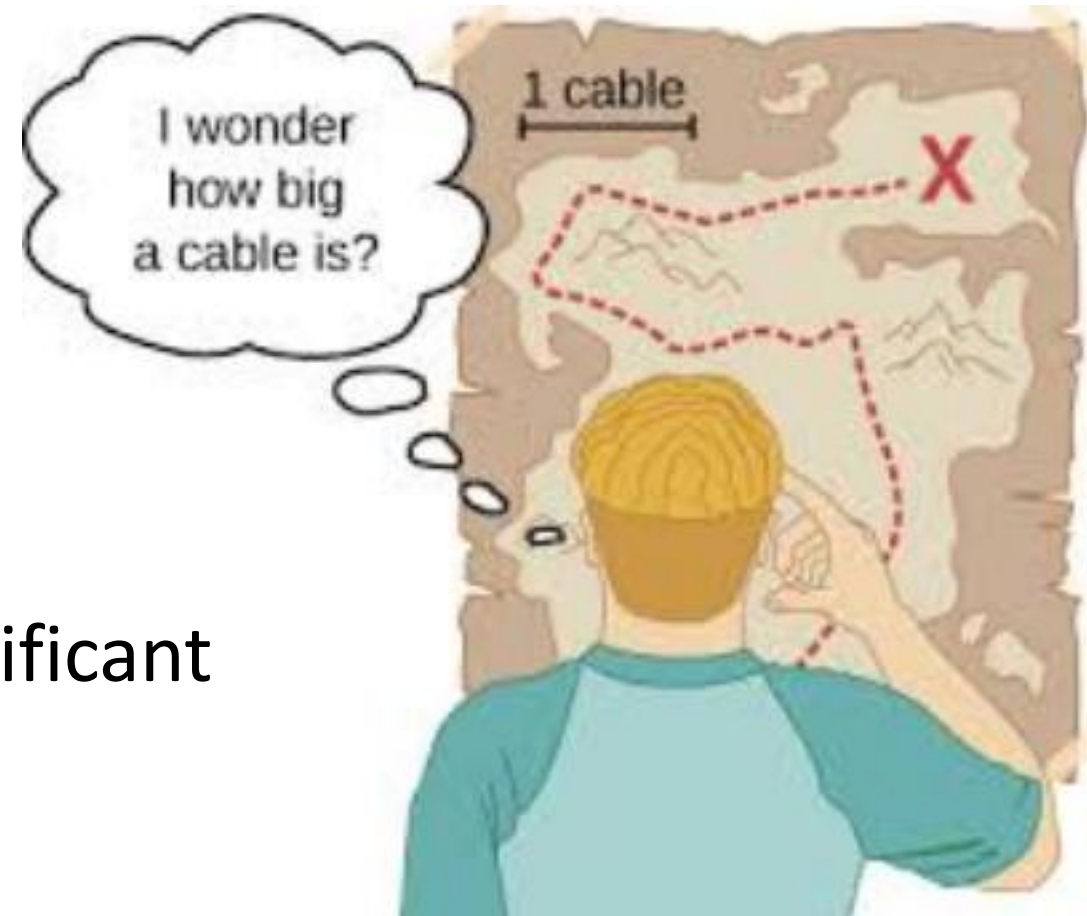
1.1 Introduction

1.2 Physical quantities and units;

- SI system,
- Unit Conversion,
- Dimensional analysis

1.3 Measurement; Uncertainty, Significant Figures

1.4 Math Review; Vectors



CHAPTER-OPENING QUESTIONS

1. How many cm^3 are in 1.0 m^3 ?

- (a) 10.
- (b) 100.
- (c) 1000.
- (d) 10,000.
- (e) 100,000.
- (f) 1,000,000.

2. Which of the following quantity and unit pair is not considered as SI (Système International)?

- (a) Length --> meter
- (b) Mass --> kilogram
- (c) Time --> second
- (d) Electric current --> ampere
- (e) Temperature --> celsius
- (f) Luminuous intenity --> candela

1.2 Physical quantities and units; SI system

Learning Outcomes

Students should be able to:

- recognize and differentiate between various **physical quantities** like length, mass, time, temperature, electric current, etc.
- use the correct **SI units** for different physical quantities and be able to convert between different units within the SI system (e.g., meters to kilometers, grams to kilograms).
- distinguish between **base units** (like meter, kilogram, second) and **derived units** (like meters per second for speed).

1.2 Physical quantities and units; SI system

How fast does light travel ?

How much do you weight ?

What is the radius of the Earth?

What temperature does ice melts at?

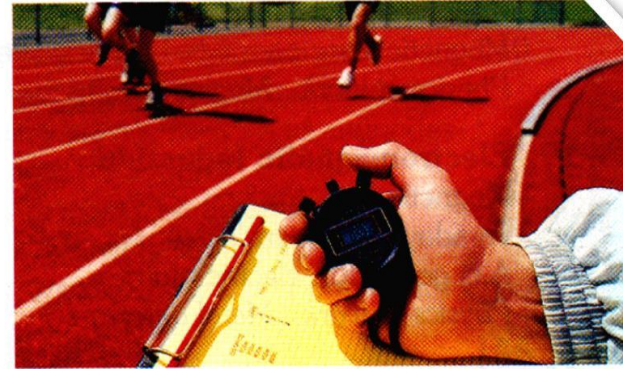
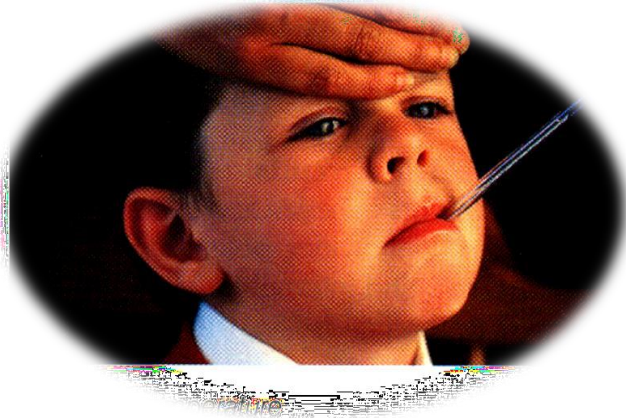
Measurement of physical quantities is an essential part of Physics.



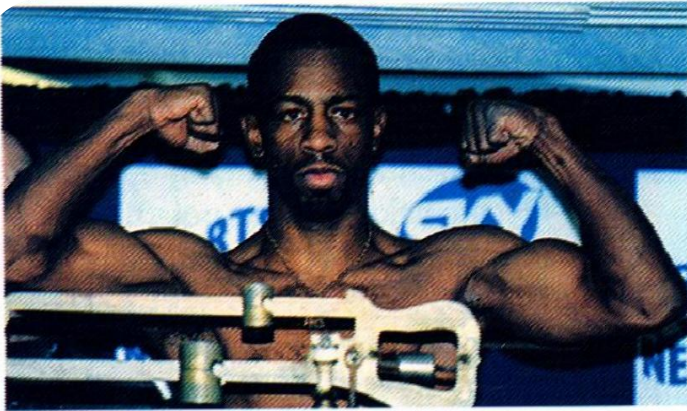
We can find the answers to all of these questions by measurement.

Speed, mass, length and temperature are all examples of **physical quantities.**

Measurement of Physical Quantities...



... and time ...



... and weight

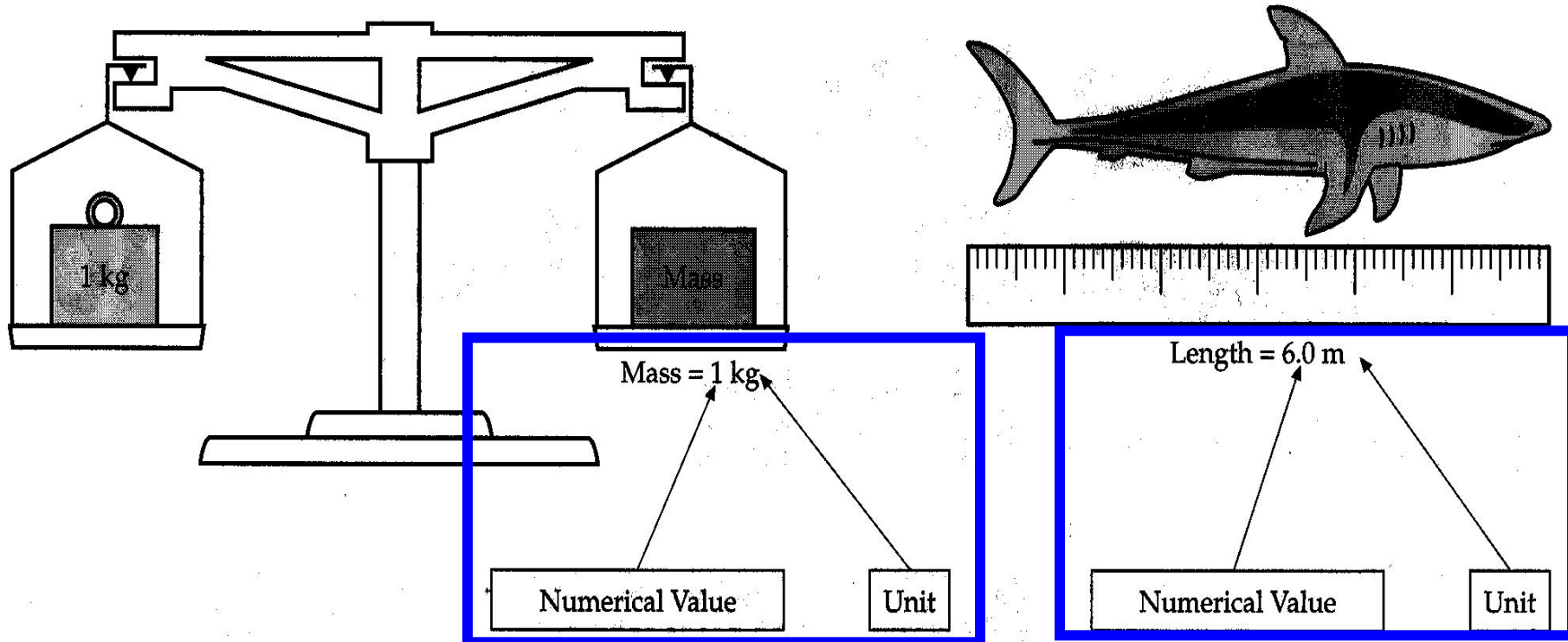


SCIENCEPHOTOGRAPHY

1.2 Physical Quantities and Units

Physical Quantities

- Physical quantity is defined as a quantity which can be measured.
- Consist of a precise numerical value & a unit.



- It can be categorized into 2 types :

1. Basic (base) Quantities

2. Derived Quantities

Physical Unit

- are standards size of measurement of physical quantities.
- ex: metre (m) – unit for length
second (s) – unit for time
Kelvin (K) – unit for temperature
- The unit of basic quantity is called base unit.
- The unit of derived quantity is called derived unit.

SI Unit

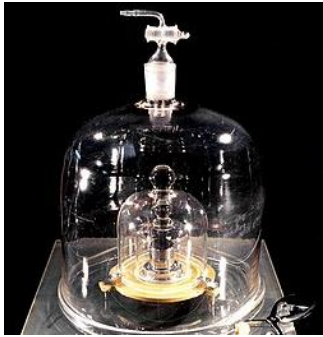
- International System of Units (Système International)
- It has been agreed internationally.

Basic (Base) Quantities & units

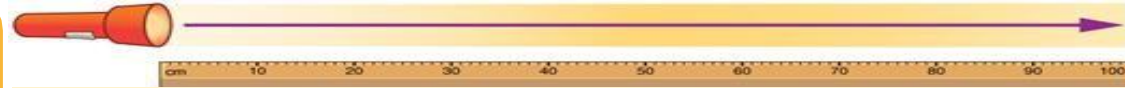
- **Basic quantity** is defined as a quantity which CANNOT BE DERIVED from any physical quantities.
 - Table 1.1 shows all the basic (base) quantities.

Quantity	Symbol	SI Unit	Symbol
Length	l	metre	m
Mass	m	kilogram	kg
Time	t	second	s
Temperature	T/θ	kelvin	K
Electric current	I	ampere	A
Amount of substance	n	mole	mol
Luminous Intensity	I_v	candela	cd

Unit – a standard size of measurement of physical quantities



1 Meter (m)



the distance traveled by the light in vacuum during a time interval of $1/299\,792\,458$ second.

1 Kilogram (kg)

The mass of a specific platinum-iridium alloy cylinder kept at the International Bureau of Weights and Measures in France (1799-2019) (2019 --> in term of Planck's constant)

1 Second (s)

one second is defined as the time required for 9,192,631,770 oscillations of Cs-133 radiation

1 Kelvin (K)

$1/273.16$ of the thermodynamic temperature of the triple point of water

1 Ampere (A)

the constant current which, if maintained in two straight parallel conductors of infinite length, of negligible cross section, and placed one metre apart in vacuum, would provide between the conductors a force equal to 2×10^{-7} Newton per meter per length

1 Mole (mol)

The amount of substance containing many identical units as there are atoms in 12 grams of carbon-12.

- **Derived quantities** are the physical quantities other than the base quantities.
- Derived quantity are quantity which can be expressed in term of base quantity.

OR

- Derived Quantity is defined as derived from base quantities according to a defining equation.

Examples of derived quantitiesVelocity, v

$$\frac{\text{length}}{\text{time}} = \frac{l}{t} = \frac{\text{m}}{\text{s}} = \text{m s}^{-1}$$

Force, F

$$\begin{aligned} &= \text{mass} \times \text{acceleration} \\ &= m \times a \\ &= \text{kg} \times \text{m s}^{-2} \\ &= \text{kg m s}^{-2} @ \text{N} \end{aligned}$$

Density, ρ

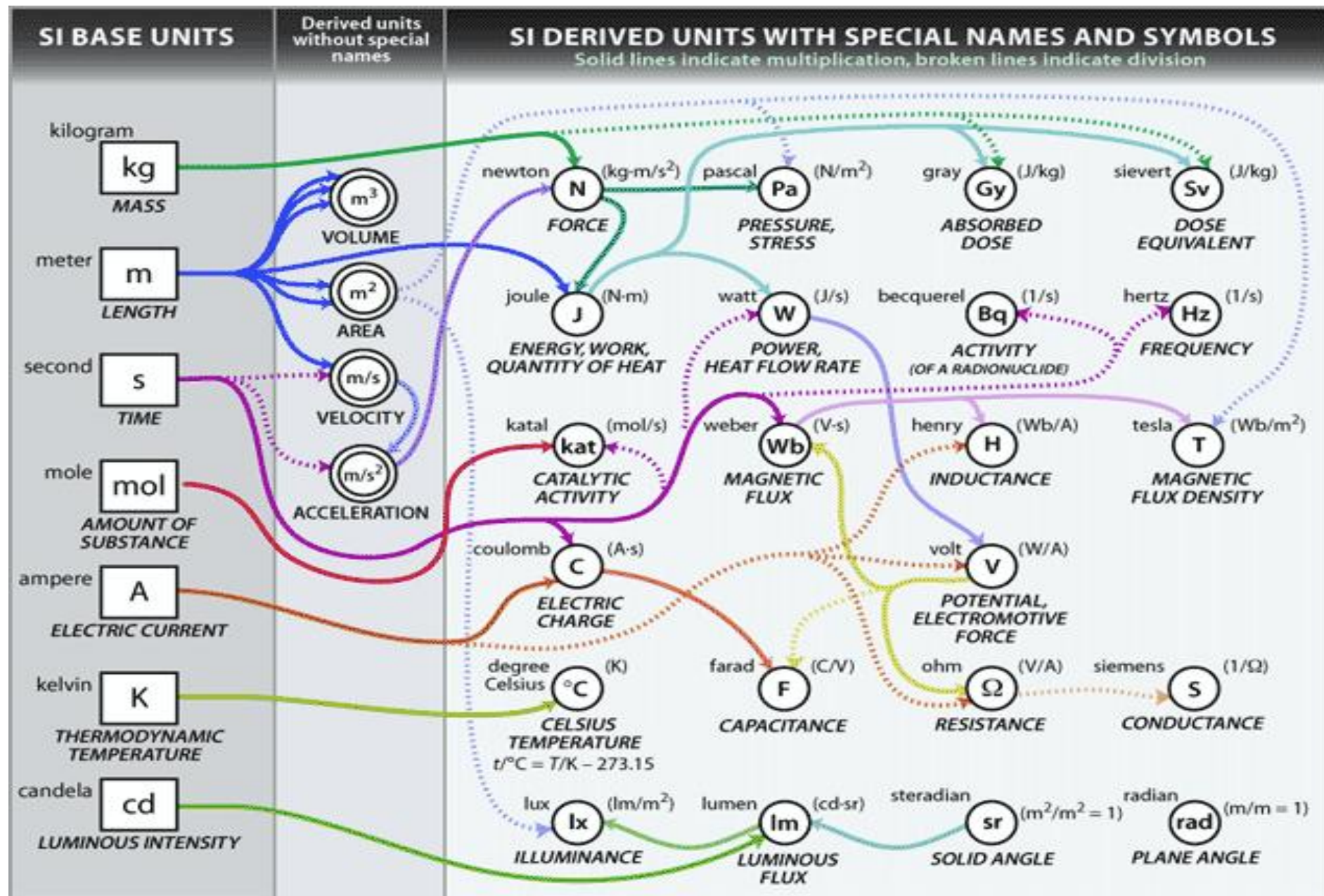
$$\begin{aligned} \frac{\text{mass}}{\text{volume}} &= \frac{m}{V} \\ &= \frac{\text{kg}}{\text{m}^3} \\ &= \text{kg m}^{-3} \end{aligned}$$

SI Unit**Special
Name**

- Table 1.2 shows some examples of derived quantity

Derived quantity	Symbol	Formulae	Unit
Velocity	v	l / t	m s^{-1}
Volume	V	$l \times w \times t$	m^3
Acceleration	a	v / t	m s^{-2}
Density	ρ	m / V	kg m^{-3}
Momentum	p	$m \times v$	kg m s^{-1}
Force	F	$m \times a$	kg m s^{-2} @ N
Work	W	$F \times s$	$\text{kg m}^2 \text{s}^{-2}$ @ J
Pressure	P	F / A	N m^{-2} @ Pa
Frequency	f	$1 / T$	s^{-1} @ Hz
Power	P	W / t	$\text{kg m}^2 \text{s}^{-3}$ @ Js^{-1} @W

Basic & Derived Quantities



- Table 1.3 shows other common units.

Quantity	Other Unit & Conversion factor	
Length	1 Angstrom ($\text{\AA}$) $\rightarrow 1 \times 10^{-10}$ meter 1 inch (in) $\rightarrow 0.0254$ meter 1 span $\rightarrow 0.2286$ meter 1 foot (ft) $\rightarrow 0.3048$ meter	1 yard (yd) $\rightarrow 0.9144$ meter 1 fathom $\rightarrow 1.8288$ meter 1 mile (mi) $\rightarrow 1609.34$ meter
Mass	1 tonne (t) $\rightarrow 1000$ kg 1 pound (lb) $\rightarrow 0.453592$ kg 1 ounce (oz) $\rightarrow 0.0283495$ kg	
Time	1 minute (min) $\rightarrow 60$ sec 1 hour (h) $\rightarrow 3600$ sec 1 day $\rightarrow 86400$ sec	1 week $\rightarrow 604800$ sec 1 month $\rightarrow 2.628 \times 10^6$ sec 1 year $\rightarrow 3.154 \times 10^7$ sec
Temperature	1 Celsius ($^{\circ}\text{C}$) $\rightarrow 273.15$ K 1 Fahrenheit ($^{\circ}\text{F}$) $= (1^{\circ}\text{F} - 32) \times 5/9 + 273.15 = 255.928$ K 1 Kelvin (K) $= (1\text{K} - 273.15) \times 9/5 + 32 = -457.9$ $^{\circ}\text{F}$	

1.2 Physical quantities and units; Unit Conversion

Learning Outcomes

Students should be able to:

- use **scientific notation and prefixes** (kilo-, milli-, etc.) to express very large or very small quantities in a concise manner.
- **Solve application problems** using metric/standard units.

1.2 Physical quantities and units; Unit Conversion

Unit Prefixes

- It is used for **presenting larger and smaller values.**
- Table 1.3 shows all the unit prefixes.

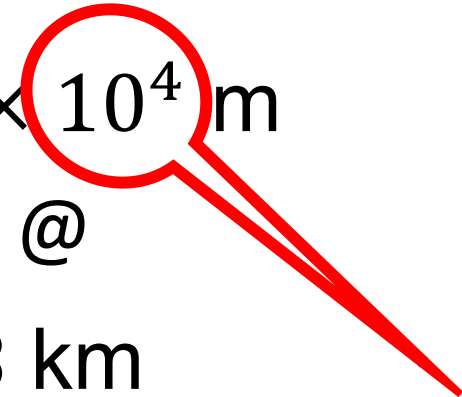


Prefix	Multiple	Numerical value	Symbol
exa	$\times 10^{18}$	1 000 000 000 000 000 000 000	E
peta	$\times 10^{15}$	1 000 000 000 000 000 000	P
tera	$\times 10^{12}$	1 000 000 000 000 000	T
giga	$\times 10^9$	1 000 000 000	G
mega	$\times 10^6$	1 000 000	M
kilo	$\times 10^3$	1 000	k
deci	$\times 10^{-1}$	0.1	d
centi	$\times 10^{-2}$	0.01	c
milli	$\times 10^{-3}$	0.001	m
micro	$\times 10^{-6}$	0.000 001	μ
nano	$\times 10^{-9}$	0.000 000 001	n
pico	$\times 10^{-12}$	0.000 000 000 001	p
femto	$\times 10^{-15}$	0.000 000 000 000 001	f
atto	$\times 10^{-18}$	0.000 000 000 000 000 001	a



Example of unit prefixes

Example 1

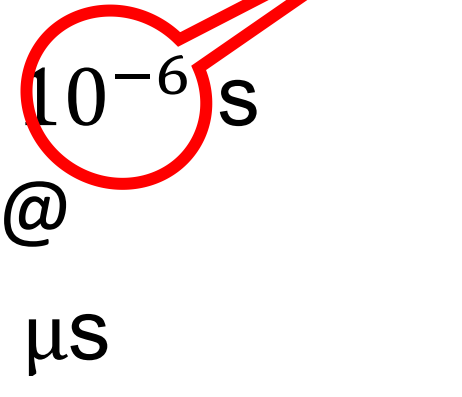
$$58000 \text{ m} = 5.8 \times 10^4 \text{ m}$$


@

58 km

**Scientific
notation**

Example 2

$$0.000009 \text{ s} = 9 \times 10^{-6} \text{ s}$$


@

9 μs

Example of unit prefixes

Example 3

Wavelength of an X-Ray = 0.000 000 001 m


$$= 1 \times 10^{-9} \text{ m @ } 1 \text{ nm}$$

Example 4

5500 m = 5500 m


$$= 5.5 \times 10^3 \text{ m @ } 5.5 \text{ km}$$

Conversion of Unit

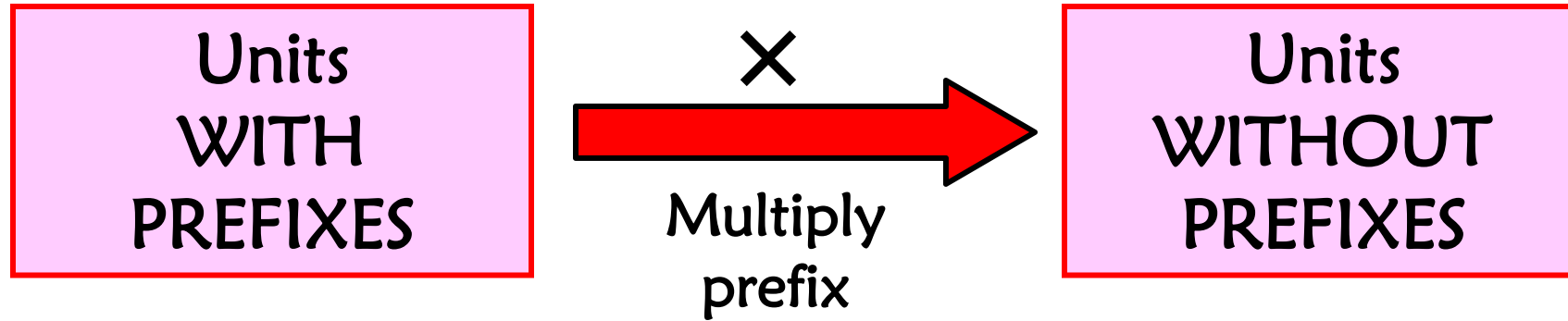
Depending on the instrument we use to measure a physical quantity, the unit may not express as an SI unit

It is often necessary to convert the unit of a physical quantity

Conversion unit let us express a quantity in terms of simpler unit **without changing its** physical value or size



Converting units WITH PREFIXES to units WITHOUT PREFIXES



Example of Conversion of Units in SI system

Example 5

1st method

$$1 \text{ cm} \rightarrow \text{m}$$

$$\text{centi} = 10^{-2}$$

$$1 \text{ cm} \Rightarrow 1 \times 10^{-2} \text{ m}$$

Example of Conversion of Units in SI system

Example 6

$$1) \ 1 \text{ cm}^2 \Rightarrow ? \text{ m}^2$$

2nd method : Cross Multiply (conversion factor)

$$= 1 \cancel{\text{cm}}^2 \times \frac{1 \text{ m}}{100 \cancel{\text{cm}}} \times \frac{1 \text{ m}}{100 \cancel{\text{cm}}}$$

$$= \frac{1 \text{ m}^2}{10000}$$

Note :
1 m = 100 cm

$$= 0.0001 \text{ m}^2 = 1 \times 10^{-4} \text{ m}^2$$

Example of Conversion of Units in SI system

Example 6

$$1 \text{ cm}^2 \Rightarrow ? \text{ m}^2$$

3rd method :

$$(\text{power})(L - R)$$

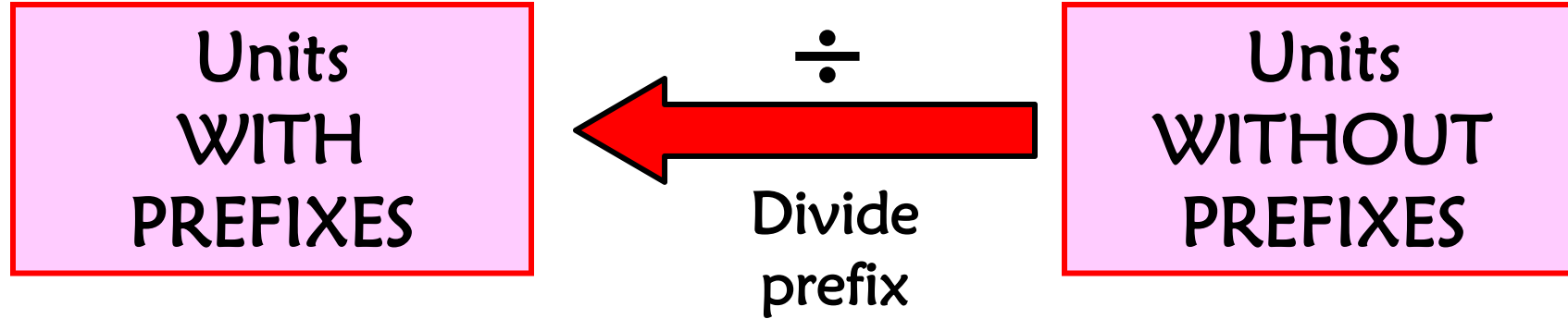
$$(2)(-2 - 0) = -4$$

$$\therefore 1 \text{ cm}^2 \Rightarrow 1 \times 10^{-4} \text{ m}^2$$

Note :

$$1 \text{ m} = 100 \text{ cm}$$

Converting units WITHOUT PREFIXES to units WITH PREFIXES



Example of Conversion of Units in SI system

Example 7

1st method :

$$1 \text{ g} \rightarrow \text{kg}$$

$$\text{kilo} = 10^3$$

$$1 \text{ g} = \frac{1}{10^3} \text{ kg} = 1 \times 10^{-3} \text{ kg}$$

Example of Conversion of Units in SI system

Example 8

$$1) \ 1 \text{ g}^2 \Rightarrow ? \text{ kg}^2$$

2nd method : Cross Multiply (conversion factor)

$$= 1 \cancel{\text{g}^2} \times \frac{1 \text{ kg}}{1000 \cancel{\text{g}}} \times \frac{1 \text{ kg}}{1000 \cancel{\text{g}}}$$

$$= \frac{1 \text{ kg}^2}{1000000}$$

Note :
1 kg = 1000 g

$$= 0.000001 \text{ kg}^2 = 1 \times 10^{-6} \text{ kg}^2$$

Example of Conversion of Units in SI system

Example 8

$$1 \text{ g}^2 \Rightarrow ? \text{ kg}^2$$

3rd method :

$$(\text{power})(L - R)$$

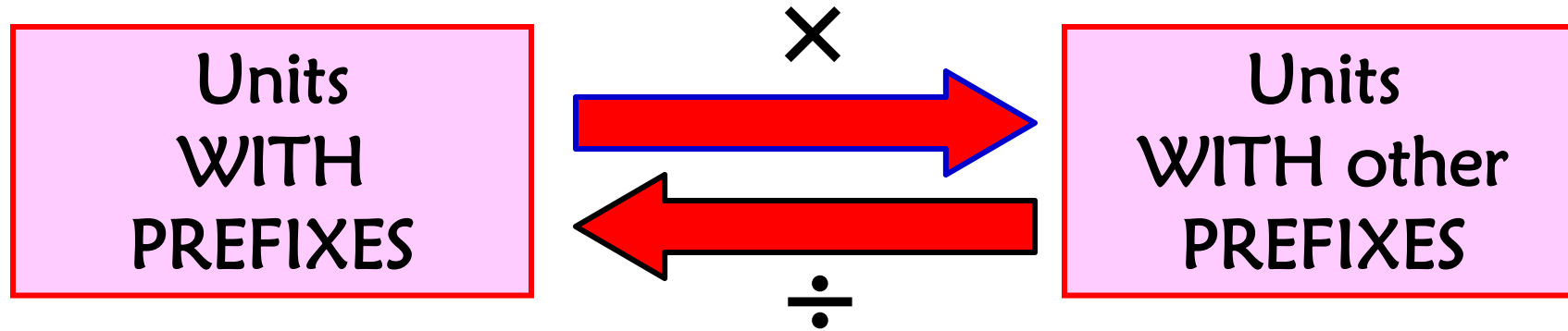
$$(2)(0 - 3) = -6$$

$$\therefore 1 \text{ g}^2 \Rightarrow 1 \times 10^{-6} \text{ kg}^2$$

Note :

$$1 \text{ kg} = 1000 \text{ g}$$

Converting units WITH PREFIXES to units WITH other PREFIXES



Example of Conversion of Units in SI system

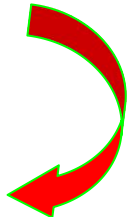
Example 9 1st method :

1 mg → g → kg

kilo = 10^3 , mili = 10^{-3}

$$1 \text{ mg} = \frac{1 \times 10^{-3}}{10^3} = 1 \times 10^{-6} \text{ kg}$$

**1st
Method**



Example of Conversion of Units in SI system**Example 10**

$$360 \text{ km h}^{-1} \Rightarrow ? \text{ m s}^{-1}$$

1st method :

$$360 \text{ km h}^{-1} = \left(\frac{360 \text{ km}}{1 \text{ h}} \right)$$

$$360 \text{ km h}^{-1} = \left(\frac{360 \times 10^3 \text{ m}}{3600 \text{ s}} \right)$$

$$= 100 \text{ m s}^{-1}$$

Note :

$$1 \text{ km} = 1000 \text{ m}$$

$$1 \text{ h} = 3600 \text{ s}$$

$$*1 \text{ min} = 60 \text{ s}$$

$$*1 \text{ h} = 60 \text{ min}$$

Example of Conversion of Units in SI system

Example 10

$$360 \text{ km h}^{-1} \Rightarrow ? \text{ m s}^{-1}$$

2nd method : Cross Multiply (conversion factor)

$$= \frac{360 \cancel{\text{km}}}{\cancel{\text{h}}} \times \frac{1000 \text{ m}}{1 \cancel{\text{km}}} \times \frac{1 \cancel{\text{h}}}{60 \cancel{\text{min}}} \times \frac{1 \cancel{\text{min}}}{60 \text{ s}}$$

$$= \frac{360 \times 1000 \text{ m}}{3600 \text{ s}}$$

$$= 100 \text{ ms}^{-1}$$

Note :

$$1 \text{ km} = 1000 \text{ m}$$

$$1 \text{ h} = 3600 \text{ s}$$

$$*1 \text{ min} = 60 \text{ s}$$

$$*1 \text{ h} = 60 \text{ min}$$

Example of Conversion of Units in SI system

$$15 \text{ km h}^{-1} =$$

$$5 \text{ mm}^3 =$$

$$7 \text{ g cm}^{-3} =$$

$$30 \text{ }^{\circ}\text{C} =$$



Every answer **must followed with unit of that quantity otherwise mark will be deducted in an examination.**

Follow Up Exercise

1. A hall bulletin board has an area of 250 cm^2 . What is this area in square meters (m^2) ?
2. The density of metal mercury is 13.6 g/cm^3 . What is this density as expressed in kg/m^3
3. A sheet of paper has length 27.95 cm , width 8.5 cm and thickness of 0.10 mm . What is the volume of a sheet of paper in m^3 ?
4. Convert the following into its SI unit:
 - (a) $80 \text{ km h}^{-1} = ? \text{ m s}^{-1}$
 - (b) $450 \text{ g cm}^{-3} = ? \text{ kg m}^{-3}$
 - (c) $15 \text{ dm}^3 = ? \text{ m}^3$
 - (d) $50^\circ \text{ C} = ? \text{ K}$

1.2 Physical quantities and units; Dimensional analysis

Learning Outcomes

Students should be able to:

- **differentiate between dimensions** (e.g., mass, length, time) **and units** (e.g., kilograms, meters, seconds).
- **distinguish between fundamental dimensions** (like those mentioned above) and dimensions that are **derived** from them (e.g., velocity, acceleration, force).
- use dimensional analysis to **perform unit conversions accurately**.
- use dimensional analysis to **check if the equations are dimensionally consistent/homogeneous** (i.e., all terms have the same dimensions).

1.2 Physical quantities and units; Dimensions and Dimensional Analysis

- Dimensions of a quantity are the base units that make it up; they are generally written using square brackets.
- Example: Speed = distance / time
- Dimensions of speed: $[L/T]$

Quantity	Dimension	SI unit
speed or velocity	$[L/T]$	m/sec
acceleration	$[L/T^2]$	m/sec^2
momentum	$[ML/T]$	kg·m/sec
force	$[ML/T^2]$	$kg·m/sec^2$
work or energy	$[ML^2/T^2]$	$kg·m^2/sec^2$
power	$[ML^2/T^3]$	$kg·m^2/sec^3$

1.2 Physical quantities and units; Dimensions and Dimensional Analysis

- Dimensions can be used as a help in working out relationships, a procedure referred to as **dimensional analysis**.
- One useful technique is the use of dimensions to **check if a relationship is *incorrect***.

For example, suppose you derived the equation $v = v_0 + \frac{1}{2}at^2$, where v is the speed of an object after a time t , v_0 is the object's initial speed, and the object undergoes an acceleration a .

$$\left[\frac{L}{T} \right] \stackrel{?}{=} \left[\frac{L}{T} \right] + \left[\frac{L}{T^2} \right] [T^2]$$

$$\stackrel{?}{=} \left[\frac{L}{T} \right] + [L].$$

The dimensions are incorrect: on the right side, we have the sum of quantities whose dimensions are not the same.

An error was made in the derivation of the original equation!!!

1.2 Physical quantities and units; Dimensions and Dimensional Analysis

→ Dimensional analysis can also be used as a **quick check on an equation** you are not sure about!!!

For example, consider a simple pendulum situation.
Suppose that you can't remember whether the equation for the period T , (the time to make one back-and-forth swing)

$$T = 2\pi\sqrt{\ell/g} \text{ or } T = 2\pi\sqrt{g/\ell},$$

$$[T] = \sqrt{\frac{[L]}{[L/T^2]}} = \sqrt{[T^2]} = [T],$$

$$[T] \neq \sqrt{\frac{[L/T^2]}{[L]}} = \sqrt{\frac{1}{[T^2]}} = \frac{1}{[T]}.$$

1.2 Physical quantities and units; Dimensions and Dimensional Analysis

Fundamental and derived SI units

- You can use units to prove that equations are invalid.

EXAMPLE: Given that distance is measured in meters, time in seconds and acceleration in meters per second squared, show that the formula $d = at$ does not work and thus is not valid.

SOLUTION: Start with the formula, then substitute the units on each side. Cancel to where you can easily compare left and right sides:

$$\cancel{d = at}$$

$$m = \frac{m}{s^2} \cdot s$$

$$m \neq \frac{m}{s}$$

FYI

- The last line shows that the units are inconsistent on left and right.
- Thus the equation cannot be valid.

1.2 Physical quantities and units; Dimensions and Dimensional Analysis

Fundamental and derived SI units

- You can use units to prove that equations are invalid.

PRACTICE: Decide if the formulas are dimensionally consistent. The information you need is that v is measured in m/s, a is in m/s², x is in m and t is in s.

(a) $v = at^2$

Inconsistent

(b) $v^2 = 3ax$

Consistent

(c) $x = at^2$

Consistent

← numbers don't have units

FYI

- The process of substituting units into formulas to check for consistency is called **dimensional analysis (DA)**.
- DA can be used only to show the invalidity of a formula. Both (b) and (c) are consistent but neither is correct. They should be: $v^2 = v_0^2 + 2a(\Delta x)$ and $x = x_0 + v_0t + (1/2)at^2$.

1.3 Measurement; Uncertainty, Significant Figures

Learning Outcomes

Students should be able to:

- understand that significant figures reflect the **precision of a measurement**.
- know the rules for determining the number of significant figures in each number, including those **relating to zeros and decimal points**.
- apply the rules for significant figures when **performing calculations** (addition, subtraction, multiplication, division) to ensure that **the final answer reflects the precision** of the input values.

1.3 Measurement; Uncertainty, Significant Figures

Introduction

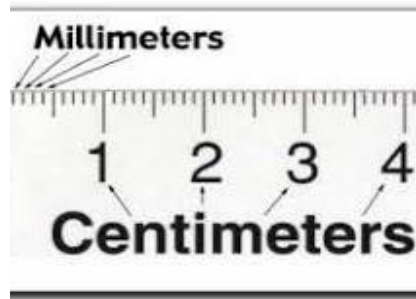
- The accuracy of a measurement depends on:
 - the sensitivity of apparatus,
 - the skill of person carrying out the measurement,
 - the number of times the measurements is repeated.



Subjects	Time/Condition		
	T ₁	T ₂	T ₃
S ₁	S ₁	S ₁	S ₁
S ₂	S ₂	S ₂	S ₂
S ₃	S ₃	S ₃	S ₃
S ₄	S ₄	S ₄	S ₄
S ₅	S ₅	S ₅	S ₅
S ₆	S ₆	S ₆	S ₆

Uncertainty in measurement

- No measurement is exact; there is always some uncertainty due to limited instrument accuracy and difficulty reading results.
- If you were to use a centimetre ruler to measure the width of a board, the result could be claimed to be precise to about 0.1 cm (1 mm), the smallest division on the ruler.



Measurement the width of a board with a centimeter ruler.

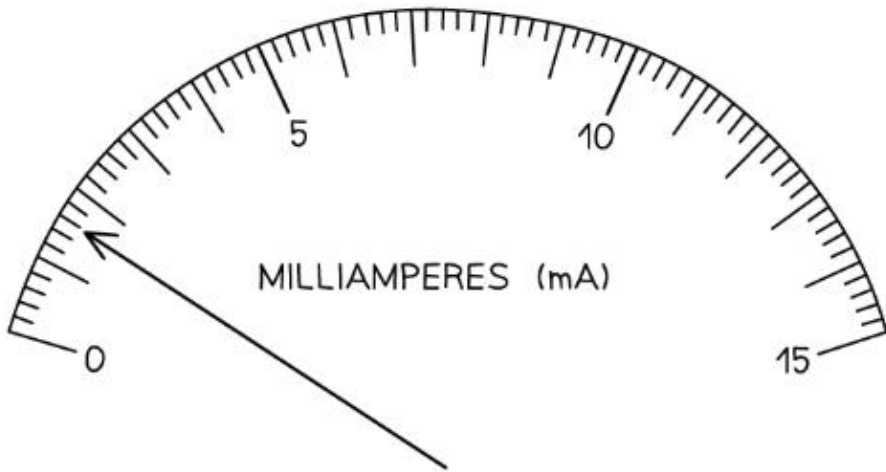
- **Estimated uncertainty** is written with a $\pm$ sign;
for example: **8.8 ± 0.1 cm**
- **Percent uncertainty** is the ratio
of the uncertainty to the measured value,
multiplied by 100:

$$\frac{0.1}{8.8} \times 100\% \approx 1\%$$

1.3 Measurement; Uncertainty, Significant Figures

Uncertainty within Measuring Device

Diagram showing a reading from an ammeter



Smallest division = 0.2 mA
Reading (I) = 1.6 mA

$$\text{Absolute Uncertainty, } (\Delta I) = \frac{1}{2} \times 0.2 \text{ mA} = 0.1 \text{ mA}$$

$$I = 1.6 \pm 0.1 \text{ mA}$$

Percentage Uncertainty,

$$(\%) = \frac{\text{Uncertainty}}{\text{Value}} \times 100 = \frac{0.1}{1.6} \times 100 = 6.2\%$$

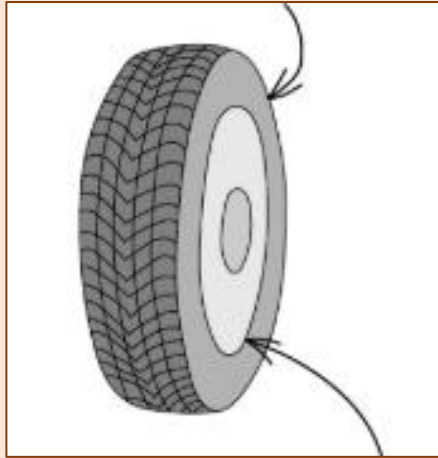
$$I = 1.6 \pm 6.2\% \text{ mA}$$

1.3 Measurement; Uncertainty, Significant Figures

Calculation Related to Uncertainty

Adding/subtracting

Diameter of outer tyre (d_1) = 55.0 ± 0.5 cm



Diameter of inner tyre (d_2) = 21.0 ± 0.7 cm

Difference in diameters ($d_1 - d_2$) = $55.0 - 21.0 = 34.0$ cm

Uncertainty in difference = $\pm(0.5 + 0.7)$ cm = ± 1.2 cm

$$\Delta d = (34.0 \pm 1.2) \text{ cm}$$

Multiply/Dividing



Distance = 50.0 ± 0.1 m

Time = 5.00 ± 0.05 s

$$\text{Speed } (v) = \frac{\text{Distance } (s)}{\text{Time } (t)}$$

$$\text{Speed } (v) = \frac{50.00 \text{ m}}{5.0} = 10.0 \text{ ms}^{-1}$$

$$\frac{\Delta v}{v} = \frac{\Delta s}{s} + \frac{\Delta t}{t} = \frac{0.1}{50.0} + \frac{0.05}{5.00} = 0.002 + 0.01 = 0.012$$

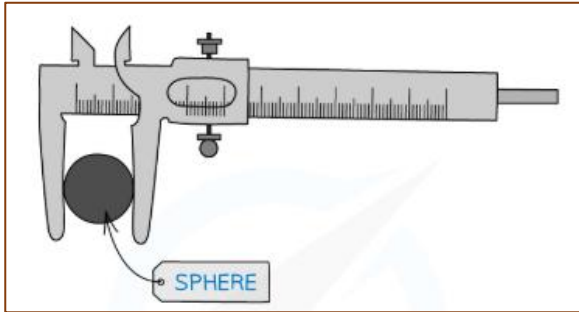
$$\text{Absolute Uncertainty, } \Delta v = 0.012 \cdot (10.0 \text{ ms}^{-1}) = 0.12 \text{ ms}^{-1}$$

$$v = (10.0 \pm 0.1) \text{ ms}^{-1}$$

1.3 Measurement; **Uncertainty**, Significant Figures

Calculation Related to Uncertainty

Raising to a power



$$V = \frac{4}{3}\pi r^3$$

$$r = 2.50 \pm 0.02 \text{ cm}$$

$$V = \frac{4}{3}\pi(2.50)^3 = 65.5 \text{ cm}^3$$

$$\frac{\Delta V}{V} = 3 \times \frac{\Delta r}{r} = 3 \times \frac{0.02}{2.50} = 0.024$$

Absolute Uncertainty,

$$\Delta V = 0.024 \cdot V = (0.024) \cdot (65.5 \text{ cm}^3) = 1.57 \text{ cm}^3$$

$$V = (65.5 \pm 1.6) \text{ cm}^3$$

Percentage Uncertainty,

$$\% \Delta V = 0.024 \cdot 100 = 2.4\%$$

$$V = (65.5 \pm 2.4\%) \text{ cm}^3$$

1.3 Measurement; Uncertainty, Significant Figures

Calculation Related to Uncertainty

Example:

Experiment: Area of a rectangular plate measurement with a meter stick.



Length:

1st reading: 16.2 cm

2nd reading: 16.3 cm

3rd reading: 16.4 cm

$$L = 16.3 \pm 0.1 \text{ cm}$$

Width:

1st reading: 4.5 cm

2nd reading: 4.4 cm

3rd reading: 4.6 cm

$$W = 4.5 \pm 0.1 \text{ cm}$$

Final area value range:

$$\text{Min: } ((16.3 - 0.01)\text{cm})((4.5 - 0.1)\text{ cm}) = 71.28 \text{ cm}^2$$

$$\text{Max: } ((16.3 + 0.01)\text{cm})((4.5 + 0.1)\text{ cm}) = 75.44 \text{ cm}^2$$

$$\text{Area Uncertainty} = \frac{(\text{Max} - \text{Min})}{2} = \frac{(75.44 - 71.28)}{2} = 2.08 \text{ cm}^2$$

$$\text{Area} = (16.3 \text{ cm})(4.5 \text{ cm}) = 73.35 \text{ cm}^2 = 73 \text{ cm}^2$$

$$\text{Area} = (73 \pm 2) \text{ cm}^2$$

Significant Figures

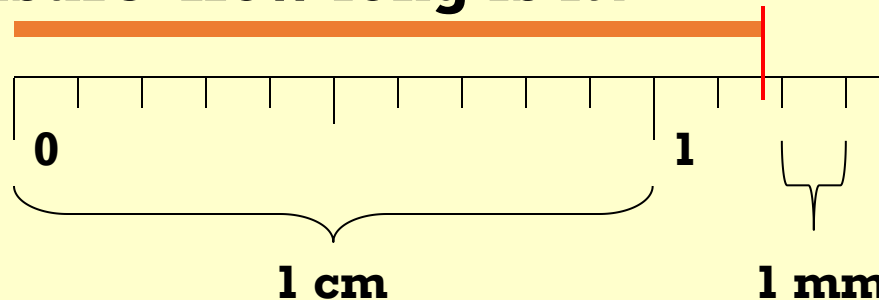
The number of **significant figures** is the number of **reliably known digits in** a number. The *way the number is written* should *indicate* the number of *significant figures*:

- 23.21 cm has 4 significant figures
- 0.062 cm has 2 significant figures (the initial zeroes don't count)
- 80 km is ambiguous—it could have 1 or 2 significant figures. If it has 3, it should be written 80.0 km.

Significant Figures

- Uncertainty in measurement is expected because of the imperfect nature of our measuring devices.
- A typical meter stick has marks at every millimeter (10^{-3} m or $1/1000$ m). Thus, the best measurement you can get from a typical meter stick is to the nearest mm.

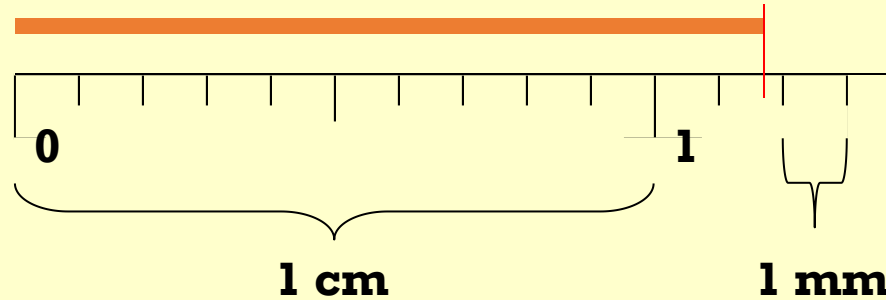
EXAMPLE: Consider the following line whose length we wish to measure. How long is it?



SOLUTION:

- It is closer to 1.2 cm than 1.1 cm, so we say it measures 1.2 cm. The 1 and 2 are both significant.

EXAMPLE: Consider the following line whose length we wish to measure. How long is it?



SOLUTION:

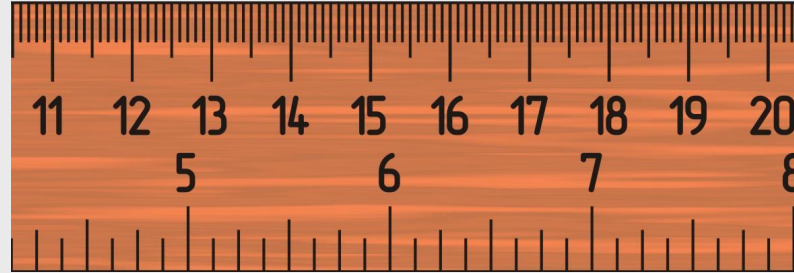
- It is closer to 1.2 cm than 1.1 cm, so we say it measures **1.2** cm.

Significant figures

- We call the “**1**” in the measurement above the **most significant digit**. It represents the “main portion” of our measurement.
- We call the “**2**” in the measurement above the **least significant digit**.

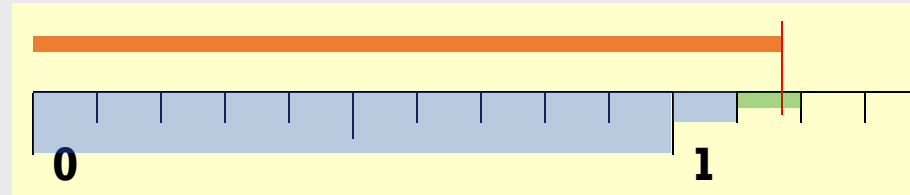
Significant figures

- A ruler is an analog measuring device.



- So is a meter with a needle.
- For good analog devices you can estimate the last digit.

- Thus, to say that the line is 1.17 cm

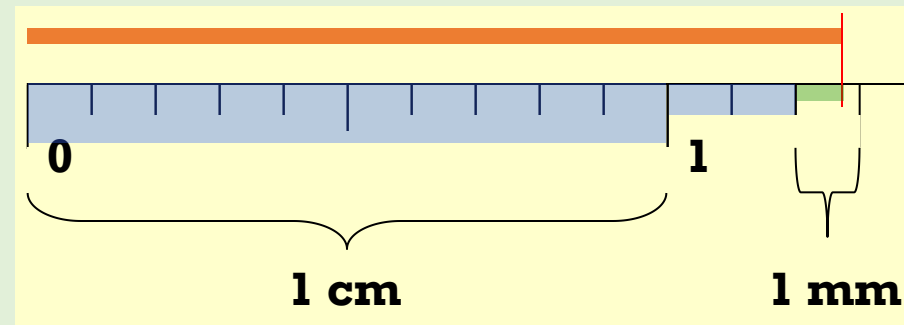


- 1.18 cm long are both correct.

- The 1.1 part constitutes the two certain digits.
- The 7 (or 8) constitutes the uncertain digit.

Estimating quantities to an appropriate number of significant figures

PRACTICE: How long is this line?

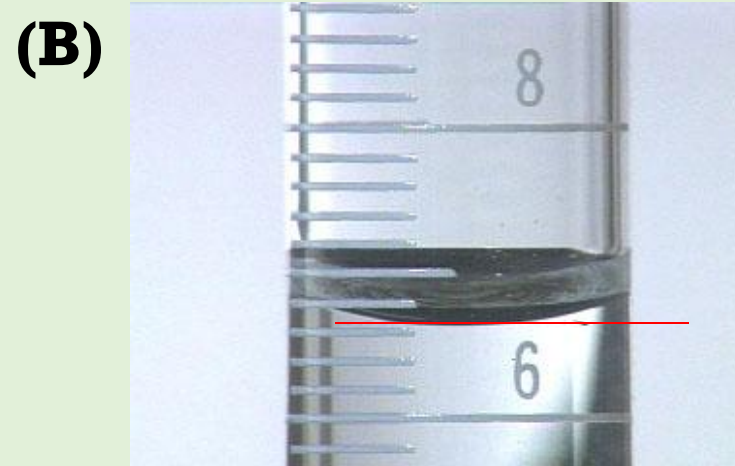
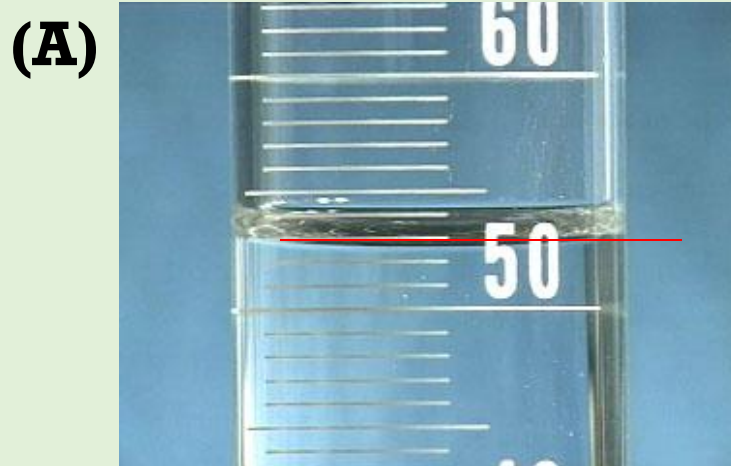


SOLUTION: Read the first two certain digits, then estimate the last uncertain one.

- The 1 and the 2 are the **certain** digits.
- The 8 (or 7) is the **uncertain** digit.
- It is about 1.28 cm (or maybe 1.27 cm) long.

Estimating quantities to an appropriate number of significant figures

PRACTICE: What is the reading on each of the graduated cylinders? Which digits are uncertain.



SOLUTION: Read to the bottom of the meniscus.

(A) reads 52.8 mL. The 8 is uncertain.

(B) Reads 6.62 mL. The 2 is uncertain.

Significant figures

- A digital measuring device, on the other hand, is only “good” to the least significant digit’s place.

EXAMPLE:

- The meter shown here is *only* good to the nearest 0.1 V. There is **NO** estimation of another digit.



Significant figures

- Significant figures are the reasonable number of digits that a measurement or calculation should have.
- For example, as illustrated before, a typical wooden meter stick has two significant figures.
- The number of significant figures in a calculation reflects the precision of the least precise of the measured values.

Rules for determining the number of Sig. Figs in measurement

Significant figures

(1) All non-zero digits are significant.	438 g	3
	26.42 m	4
	0.75 cm	2
(2) All zeros between non-zero digits are significant(captive zero)	12060 m	4
	900.43 cm	5
(3) Filler zeros of an understood decimal place are not significant.	220 L	2
	60 g	1
	30. cm	2
(4) Filler zeros (leading zero)of a decimal place are not significant.	0.006 L	1
	0.08 g	1
(5) All non-filler zeros (trailing zero) to the right of a decimal place are significant.	8.0 L	2
	60.40 g	4

Rule of Significant figures in calculations

EXAMPLE

CALCULATOR

FINAL ANS

• **Multiplication and division** – round your answer to the same number of significant digits as the quantity with the fewest number of significant digits.

$$(1.2 \text{ cm})(2 \text{ cm})$$

$$2.4 \text{ cm}^2$$

$$2 \text{ cm}^2$$

$$\pi(2.75 \text{ cm})^2$$

$$7.5625 \text{ cm}^2$$

$$7.56 \text{ cm}^2$$

$$5.350 \text{ m}/2.752 \text{ s}$$

$$1.944040698 \text{ m/s}$$

$$1.944 \text{ m/s}$$

$$(0.0075 \text{ N})(6 \text{ m})$$

$$0.045 \text{ Nm}$$

$$0.05 \text{ Nm}$$

• **Addition and subtraction** – round your answer to the same number of decimal places as the quantity with the fewest number of decimal places.

$$1.2 \text{ cm} + 2 \text{ cm}$$

$$3.2 \text{ cm}$$

$$3 \text{ cm}$$

$$2000 \text{ m} + 2.1 \text{ m}$$

$$2002.1 \text{ m}$$

$$2002 \text{ m}$$

$$0.00530 \text{ m} - 2.10 \text{ m}$$

$$-2.0947 \text{ m}$$

$$-2.09 \text{ m}$$

Calculations with sig. Figs

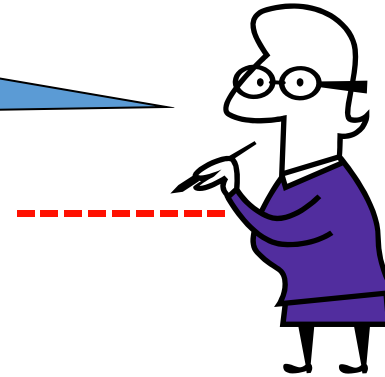
Addition and subtraction Rules:

1. If both/all number have decimal, your number can have same numbers of decimals that the number with the least has
2. For one/all numbers that does not have decimal, whatever number that have least significant places is how many significant places your answer can have
3. Math first, significant figures later

Calculations with Sig. figs.

Multiplication and division (measurements):

Check the least
significant figure!



$$2.734 \text{ cm} \times 5.2 \text{ cm} \times 8.1294 \text{ cm} =$$

4 s.f.

2 s.f.

5 s.f.

115.5740539 cm³

written as 120 cm³

2 s.f.'s

Calculations with sig. Figs.

Addition and subtraction:



Look at the least decimal place!

$$\begin{array}{rcl} 3.63 & \text{cm} & 2 \text{ d.p.} \\ 13.129 & \text{cm} & 3 \text{ d.p.} \\ +123.1 & \text{cm} & 1 \text{ d.p.} \end{array}$$

$$139.859 \text{ cm} = 139.9 \text{ cm}$$

significant to
the 0.1 place

Rules for performing a combination of operation

Example: $1.4 \times 2.639 + 117.25$ (Order of Operations $\rightarrow$ Multiply then Add)

This is **CORRECT**!

Step 1: Multiply -

$$1.4 \times 2.639 + 117.25$$

$$= 3.6946$$

Because this is multiplication, the answer to this part can only have 2 SF's. This means that the first and second digits are significant. For this particular problem, this means the ones place and the first decimal place. Make note of this, but do **NOT** round yet.

This is **NOT** Correct!

$$= 1.4 \times 2.639 + 117.25$$

$$= 3.7 + 117.25 = 120.95 \neq 121.0$$

Error - Rounded too early.
Should have been left as
3.6946.

Please be cautious!!
That you do not round the
answer too early!

Rules for performing a combination of operation

Step 2: Addition -

$$= 3.6946 + 117.25 = 120.9446 = 120.9 \text{ (Correct Answer)}$$

1 decimal place 2 decimal places

Answer needs to be rounded to 1 decimal place because that is the least precise number in the addition calculation.

Rules for performing a combination of operation

If you have both addition/subtraction and multiplication/division in a formula;
The previous rules listed for A/S and M/D still apply along with incorporating the rules for order of operations (PEMDAS)

P - Parenthesis first
E - Exponent next
M > Division and Multiplication
D > from left to right
A > Addition and Subtraction
S > from left to right

Be cautious that you do not round the answer too early!

Rules for performing a combination of operation

Exercise:

1) $3.6 \times 4.639 + 5.831$ [Ans=23]

2) $(9.24 - 1.342) / 2.3$ [Ans=3.4]

3) $(14.7 \text{ m} \times 7.5 \text{ m}) + (13.8 \text{ m}^2)$ [Ans= 124 m²]

*Order-of Magnitude: Rapid Estimating

- Precise answers are sometimes difficult to calculate, either for mathematical reasons or because limited information is available.
- Sometimes we are interested only in an approximate value.
- **Benefits:**
 - providing sufficient precision, or
 - serving as a check on more detailed calculations.

E.g.: If a good estimate shows a large answer is expected but a detailed calculation gives a small answer- calculation error.

- Useful estimates are often made by performing **order-of-magnitude** calculations in which the values are reasonably approximated to at most **one significant figure and its power of 10**.

Estimating:

A typical state has ~ 100 hospitals.

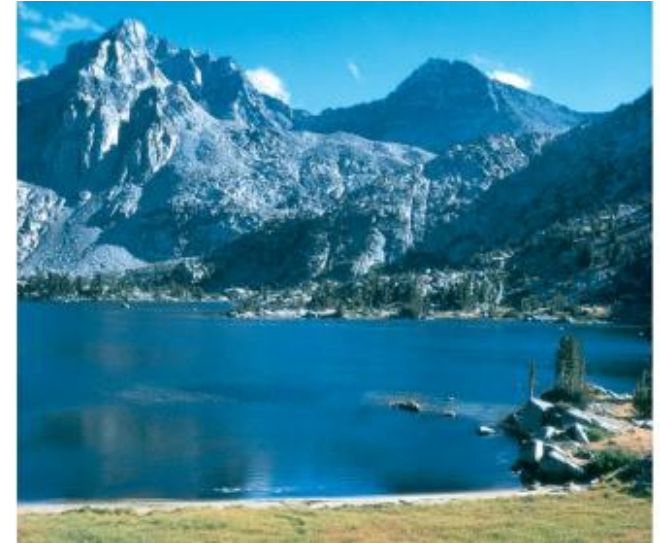
Symbol $\sim$ means “is on the order of” or “is approximately”.

*Order-of Magnitude: Rapid Estimating

Example:

Question: Estimate how much water there is in a particular lake (in Figure) which is roughly circular, about 1 km across, and you guess it has an average depth of about 10 m.

Approach: No lake is a perfect circle, nor can lakes be expected to have a perfectly flat bottom. We are only estimating here.



Solution: The volume V of a cylinder is the product of its height h times the area of its base: $V = h\pi r^2$ where r is the radius of the circular base.

Since $\pi (\pi) \approx 3.14159 \approx 3$

$$V = h\pi r^2 \approx (10 \text{ m}) \times (3) \times (5 \times 10^2 \text{ m})^2 \approx 8 \times 10^6 \text{ m}^3 \approx 10^7 \text{ m}^3$$

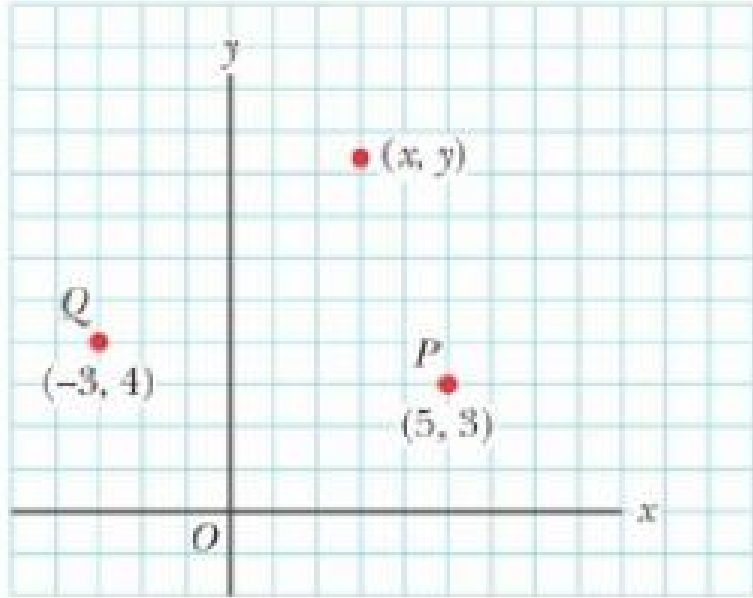
1.4 Math Review & Introduction to Vectors

Learning Outcomes

Students should be able to:

- equip with the **necessary mathematical skills** to understand and solve physics problems; coordinate system
- rearrange equations, work with **exponents and logarithms**, understand **vector operations**, and apply **trigonometric functions**

1.4 Math Review; Coordinate System & Trigonometry



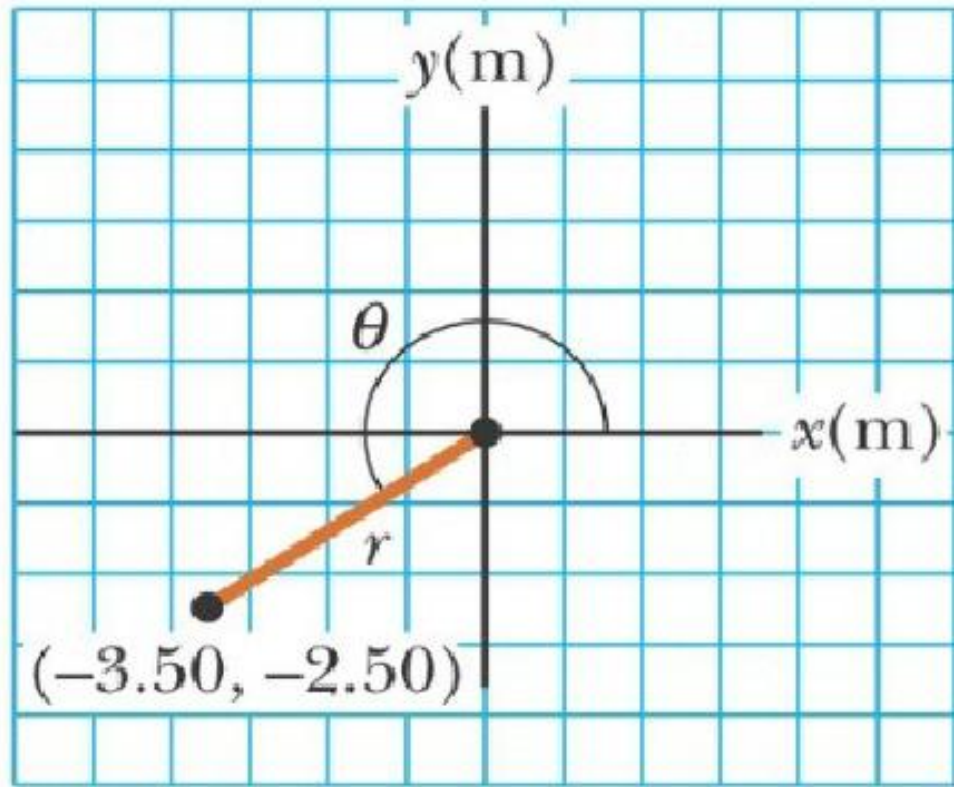
Example:

Point P: 5 m horizontally to the right,
3 m vertically upward from origin.

Point Q: 3 m horizontally to the left,
4 m vertically upward from origin.

- Many aspects of physics deals with **locations in space**, which require **the definition of a coordinate system**.
- A coordinate system (Cartesian coordinate/rectangular coordinate) consists of:
 - A fixed **reference point** O , called the origin.
 - A **set of specified axes**, or directions, with an *appropriate scale and labels* on the axes.
 - Instructions on **labeling a point** in space *relative to the origin and axes*.

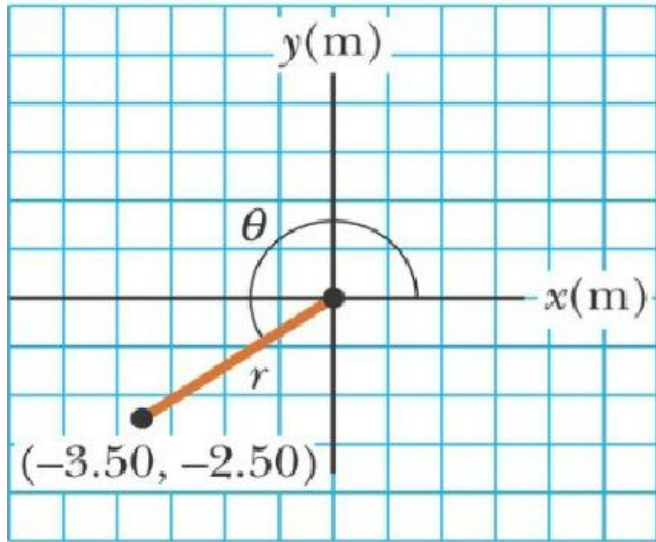
1.4 Math Review; Coordinate System & Trigonometry



- Sometimes may show as plane polar coordinate (r, θ) .
- A point is specified by the distance r from the origin to the origin and by the angle θ between **the reference line (+x)** (where the $\theta=0$) and **a line drawn from a origin to the point (red line)**.

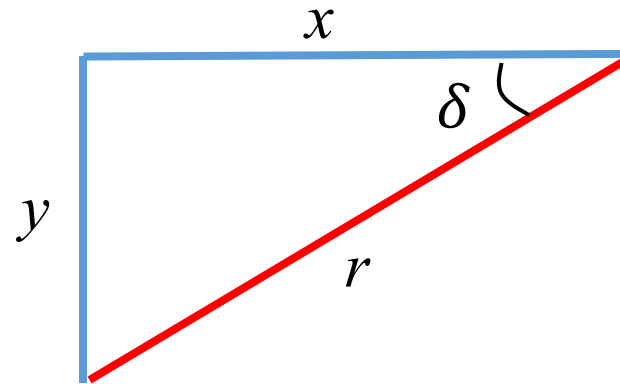
1.4 Math Review; Coordinate System & Trigonometry

Examples:



RECALL! Pythagorean Theorem and Trigonometry

$$r^2 = x^2 + y^2$$



$$\tan \delta = \frac{y}{x}$$

$$\sin \delta = \frac{y}{r}$$

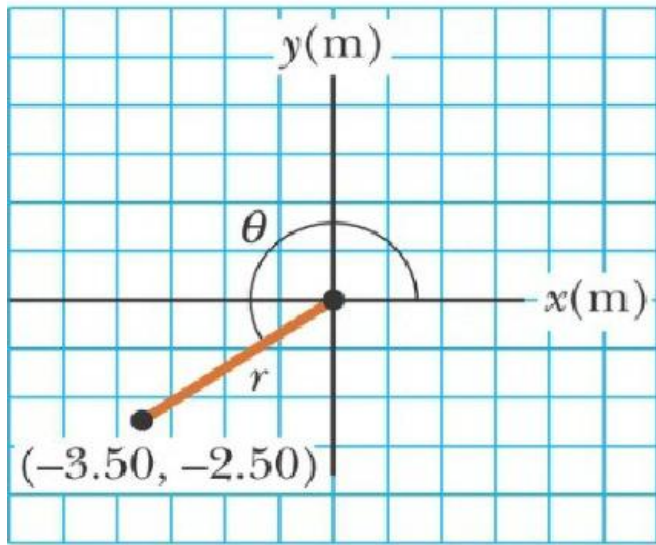
$$\cos \delta = \frac{x}{r}$$

Polar coordinate (r, θ)

- 1) How to find r ?
- 2) How to find θ ?

1.4 Math Review; Coordinate System & Trigonometry

Examples:



$$r = \sqrt{x^2 + y^2} = \sqrt{(-3.50 \text{ m})^2 + (-2.50 \text{ m})^2} = 4.30 \text{ m}$$

$$\tan \theta = \frac{y}{x} = \frac{-2.50 \text{ m}}{-3.50 \text{ m}} = 0.714$$

$$\theta = 216^\circ$$

Cartesian coordinate: $(x, y) = (-3.50 \text{ m}, -2.50 \text{ m})$

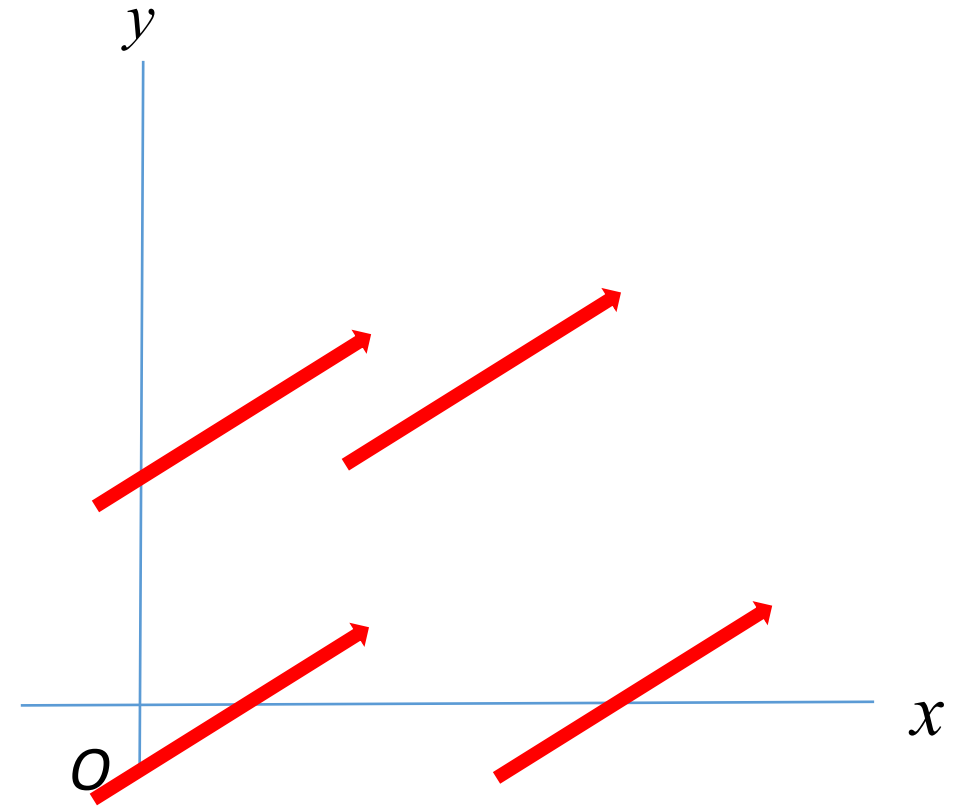
Polar coordinate: $(r, \theta) = (4.30 \text{ m}, 216^\circ)$

Since $\tan^{-1}(0.714) = 35.5^\circ$, not to forget to add 180° because the angle is actually in the 3rd quadrant.

1.4 Introduction to Vectors

Equality of vectors

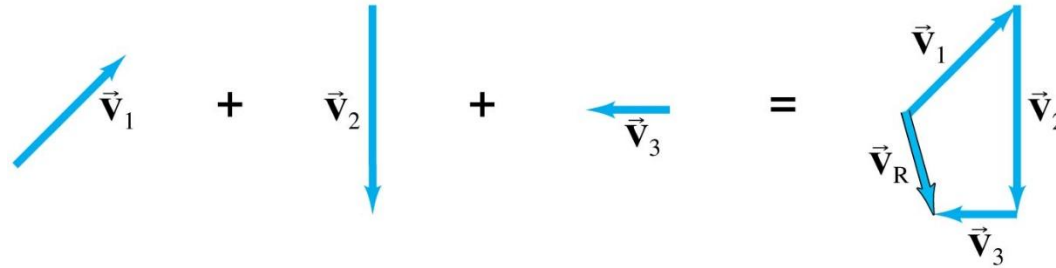
- **Scalar quantity** is just a value (magnitude only) with no direction - come with appropriate unit. (E.g.: mass, m , temperature, T , speed, v & etc.)
 - Symbols are shown in italics
- **Vector quantity** is characterized by having both a **magnitude** and a **direction**. (E.g.: velocity, $\vec{v}$, acceleration, $\vec{a}$, force, $\vec{F}$, & etc.)
 - symbols use in **boldface type**, with a **tiny arrow over the symbol**



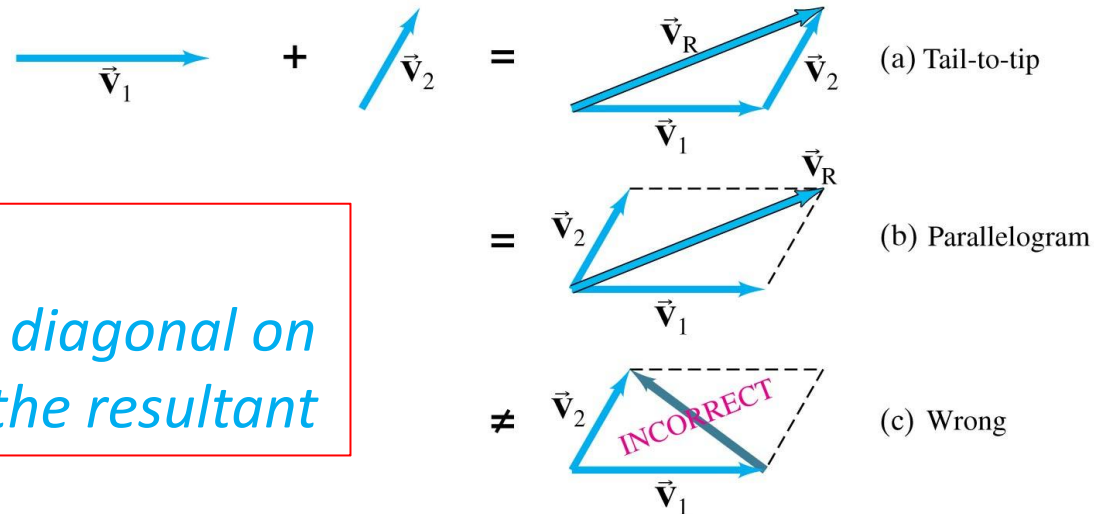
All 4 vectors are equal
(equal length & direction).

1.4 Introduction to Vectors (Vectors Addition)

Even if the vectors are not at right angles, they can be added graphically by using the “tail-to-tip” method.



The parallelogram method may also be used; “It is equivalent to tail-to-tip method”

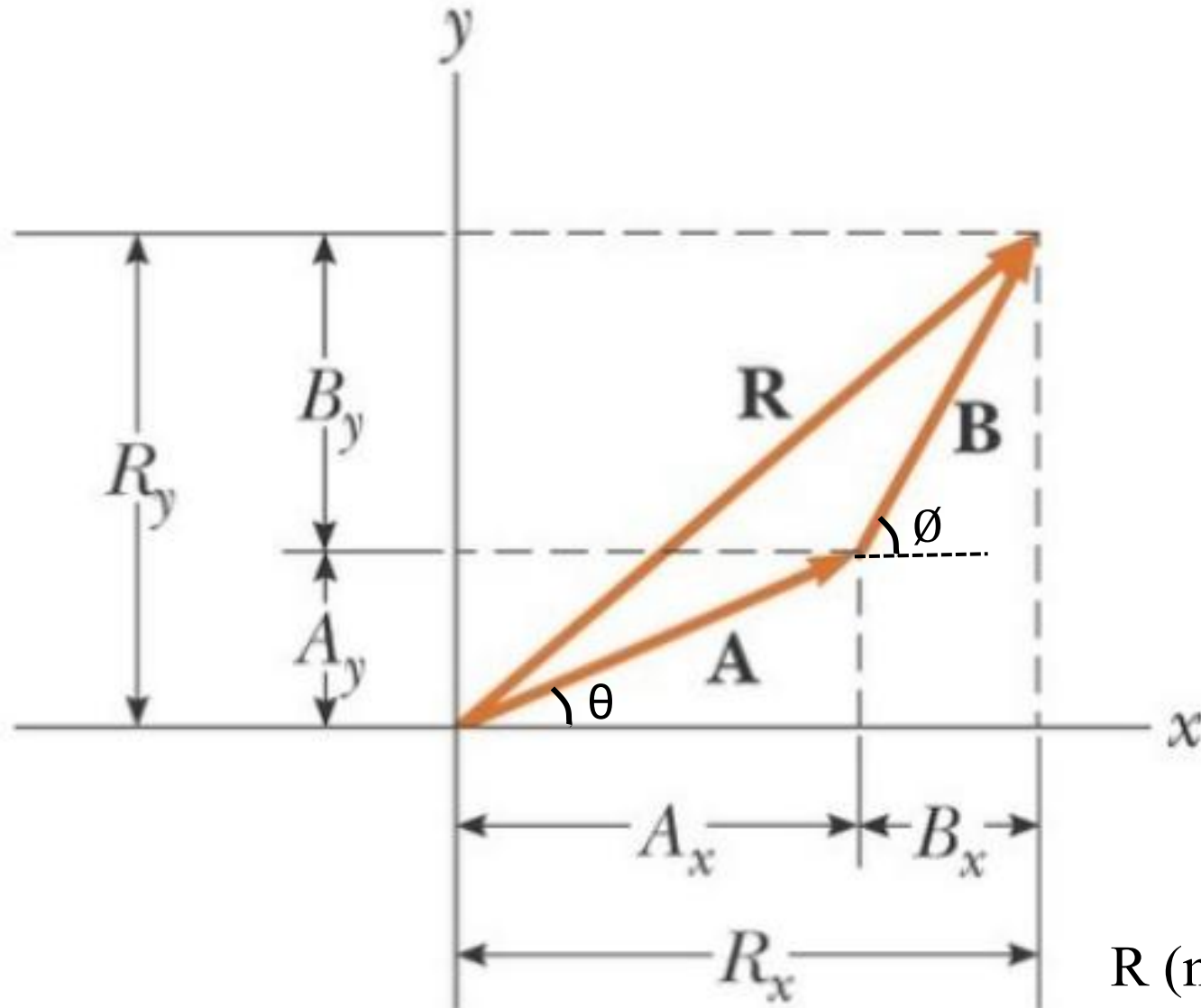


Caution!

Be sure to use the correct diagonal on the parallelogram to get the resultant

(Measurement can be done on the graph paper with ruler and protractor to obtain the magnitude and the angle of the resultant vector. But the length and the angle must be correct!)

1.4 Introduction to Vectors (Vector components)



$$\vec{A} = \vec{A}_x + \vec{A}_y$$

$$A_x = A \cos \theta$$

$$A_y = A \sin \theta$$

$$\vec{B} = \vec{B}_x + \vec{B}_y$$

$$B_x = B \cos \phi$$

$$B_y = B \sin \phi$$

$$R_x = A_x + B_x$$

$$R_y = A_y + B_y$$

R (magnitude) --> Pythagorean Theorem

R (direction) --> Trigonometry

* Problem solving Strategy

1. Read Problem - very sure you understand the nature of the problem
2. Draw Diagram - visualize the problem/situation
3. Label Physical Quantities - If involve with coordinate system (draw and label it)
4. Identify Principle(s); List data (knowns and unknowns)
5. Choose Equation - find appropriate relationship - unit guide
6. Solve Equation (s) - Substitute known values
7. Check Answer - do the units match?, is the answer reasonable?, does the plus and minus sign make sence?, is your answer consistent with an order of magnitude estimate?